



GLA UNIVERSITY

PROJECT SYNOPSIS

Submitted By

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Submitted To

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MERN STACK • FULL-STACK WEB APPLICATION

PROJECT DOCUMENTATION

Declaration

We hereby declare that the **ShopSphere — E-Commerce Platform** project will be designed and developed by our team as a mini project. The project will involve the planning, designing, development, implementation, testing, and documentation of an e-commerce platform. All the project work, including research, development, implementation, and preparation of the required documentation, will be carried out by the team members listed in this synopsis. We further declare that the work presented in this project will be completed by our team with proper guidance and coordination throughout the development process.

Team Members

S. No.	Name	ID
1	[PALAK AGARWAL]	[2415001077]
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Acknowledgement

We would like to express our sincere gratitude to our faculty members and technical trainer for their valuable guidance, continuous support, and encouragement throughout the development of our **ShopSphere — E-Commerce Platform** project. Their suggestions, feedback, and technical assistance helped us understand the various aspects of the project and complete the work successfully.

We are also thankful for the learning opportunities and resources provided to us during the development process. The guidance and support we received throughout the planning, designing, implementation, testing, and documentation stages played an important role in completing this project.

Finally, we would like to extend our heartfelt thanks to everyone who directly or indirectly contributed to the successful completion of this project.

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PROJECT OVERVIEW

Introduction

ShopSphere — E-Commerce Platform is a full-stack web application developed to provide a simple, convenient, and user-friendly online shopping experience. It allows customers to browse products across different categories, view product details, search and filter items, add products to a shopping cart, place orders, and track their purchases.

The application is built using the **MERN Stack**, consisting of **MongoDB, Express.js, React.js, and Node.js**. React.js is used to create a responsive and interactive user interface, while Node.js and Express.js handle server-side logic and provide RESTful APIs for communication between the frontend and database. MongoDB stores data related to users, products, carts, and orders.

The platform is designed with a **modular and scalable structure**, making it easier to maintain and extend with additional features. It incorporates concepts such as **user authentication, product management, CRUD operations, cart management, order processing, API integration, database connectivity, and responsive web design**.

The project also uses **Git and GitHub** for version control and collaborative development. Overall, ShopSphere demonstrates the practical application of modern full-stack development techniques and provides hands-on experience in building a complete e-commerce system from frontend to backend and database integration.

PROJECT FOUNDATION

Reason for Choosing the Project

The ShopSphere project was selected to gain practical experience in developing a modern full-stack web application.

The project provides an opportunity to understand and implement important concepts such as:

- Client-server architecture
- REST APIs
- User authentication and authorization
- Database management
- Product management
- Shopping cart functionality
- Order processing
- Git/GitHub-based collaborative development

Main Objective

The main objective of ShopSphere is to develop a **multi-category e-commerce platform** that allows customers to browse products, search and filter products, manage their shopping cart, complete checkout, place orders, and view their order status.

The system also provides administrative functionality for managing products, categories, and customer orders.

PROJECT SCOPE

The scope of the **ShopSphere — E-Commerce Platform** is to develop a web-based online shopping system that provides customers with a convenient and user-friendly platform for purchasing products online.

The system will allow users to **register and log in, browse products, search and filter products, view product details, add and manage items in the shopping cart, place orders, and track order status**. The system will also provide product and order management functionality for administrators.

The project will demonstrate the practical implementation of **MERN Stack technologies, RESTful APIs, database connectivity, authentication, CRUD operations, and responsive web design**.

The platform is designed with a **scalable and modular architecture**, allowing additional features such as online payment integration, product reviews and ratings, advanced search, notifications, offers, and improved order management to be added in the future.

Overall, the project aims to provide a basic yet complete **e-commerce solution** while demonstrating the development of a full-stack web application from frontend to backend and database integration.

Working Methodology

The system follows a client-server architecture:

1. React.js Frontend

Handles the user interface and user interactions.

2. REST API

Provides communication between the frontend and backend.

3. Node.js + Express.js

Handles server-side logic, authentication, validation, and API requests.

4. MongoDB

Stores users, products, categories, carts, and orders.

5. Response to User

The backend processes requests and sends appropriate responses to the React frontend.

ENVIRONMENT

System & Hardware Requirements

System Requirements

- **Operating System:** Windows / Linux / macOS
- **Code Editor:** Visual Studio Code
- **Runtime Environment:** Node.js and npm
- **Database:** MongoDB / MongoDB Compass
- **Version Control:** Git and GitHub
- **Web Browser:** Modern web browser (Chrome, Firefox, Edge, etc.)
- **API Testing Tool:** Postman

Hardware Requirements

- **RAM:** Minimum 4 GB
- **Processor:** Intel Core i3 or equivalent
- **Internet:** Stable internet connection
- **Recommended:** 8 GB or more RAM with SSD storage

TECHNOLOGY STACK

Frontend — React.js

React.js is used to build the interactive and responsive user interface. Major frontend functionalities include:

- Product listing and categories
- Product search and filtering
- Product details
- Shopping cart
- Checkout interface
- User account and order information
- Responsive user interface

Backend — Node.js + Express.js

Node.js and Express.js are used to develop the backend server and REST APIs. Major backend functionalities include:

- User authentication and authorization
- Product APIs
- Category APIs
- Cart management APIs
- Order management APIs
- Server-side validation
- Database communication

Database — MongoDB

MongoDB is used as the database for storing application data, including:

- Users
- Products
- Categories
- Shopping carts

- Orders

Development Tools

- Visual Studio Code
- Git
- GitHub
- Postman
- MongoDB Compass

SYSTEM COMPONENTS

Module Description System Modules

1. User Authentication

Manages user registration, login, logout, authorization, and basic profile information to provide secure access to the platform.

2. Product Management

Handles product categories, product details, pricing, search, filtering, and product availability for easy browsing.

3. Shopping Cart

Allows users to add or remove products, update quantities, and view the calculated cart total before checkout.

4. Checkout

Provides an order summary, collects delivery details, and allows users to review and confirm their purchases.

5. Order Management

Manages order creation, order history, order details, and order status, allowing users to track their purchases.

6. Admin Management

Allows administrators to manage products, categories, and customer orders, ensuring proper control of the platform.

DEPLOYMENT

Availability

1. Web-Based

ShopSphere is a web-based e-commerce application that can be accessed through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. Users can access the platform from different devices without requiring a separate desktop application.

2. Deployable

The application can be deployed on suitable cloud or web hosting platforms, making it accessible to users over the internet. The backend services can be hosted on a server, while the MongoDB database can be deployed using a cloud database service. This allows the system to support multiple users and provides flexibility for future expansion.

3. Extensible

ShopSphere is designed with a modular structure that allows new features and functionalities to be added easily. Future enhancements may include additional product categories, advanced search and filtering, multiple payment methods, user reviews and ratings, discount and coupon systems, notifications, and improved order management.

4. Installation

No dedicated desktop software installation is required to use ShopSphere. Once the application is deployed, users can simply open the web application through a supported browser and access its available features. For development purposes, the required Node.js, MongoDB, and project dependencies can be installed on the developer's system.

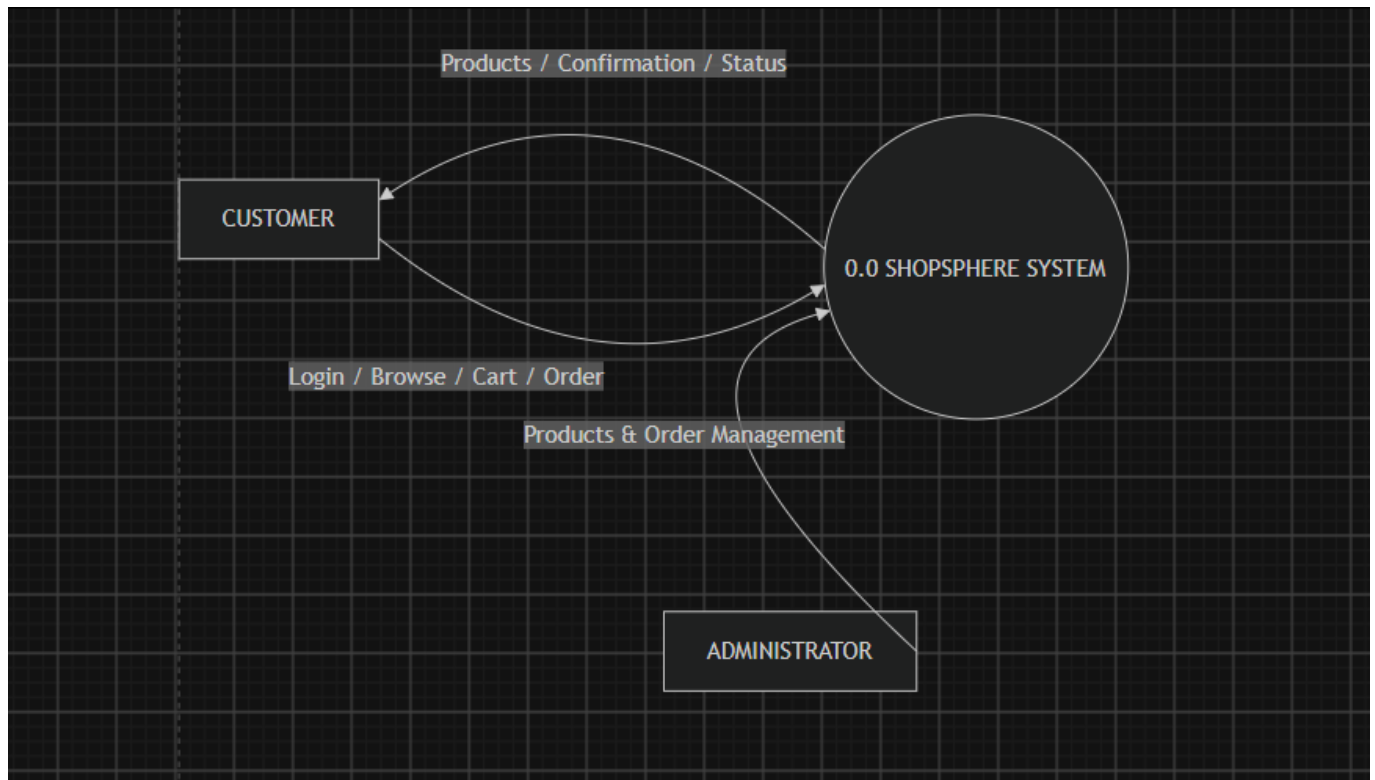
5. Cross-Platform Accessibility

Since the application runs through a web browser, it can be accessed on different operating systems, including **Windows, Linux, and macOS**. Its responsive design can also allow users to access the platform from desktops, laptops, tablets, and compatible mobile devices.

DATA FLOW

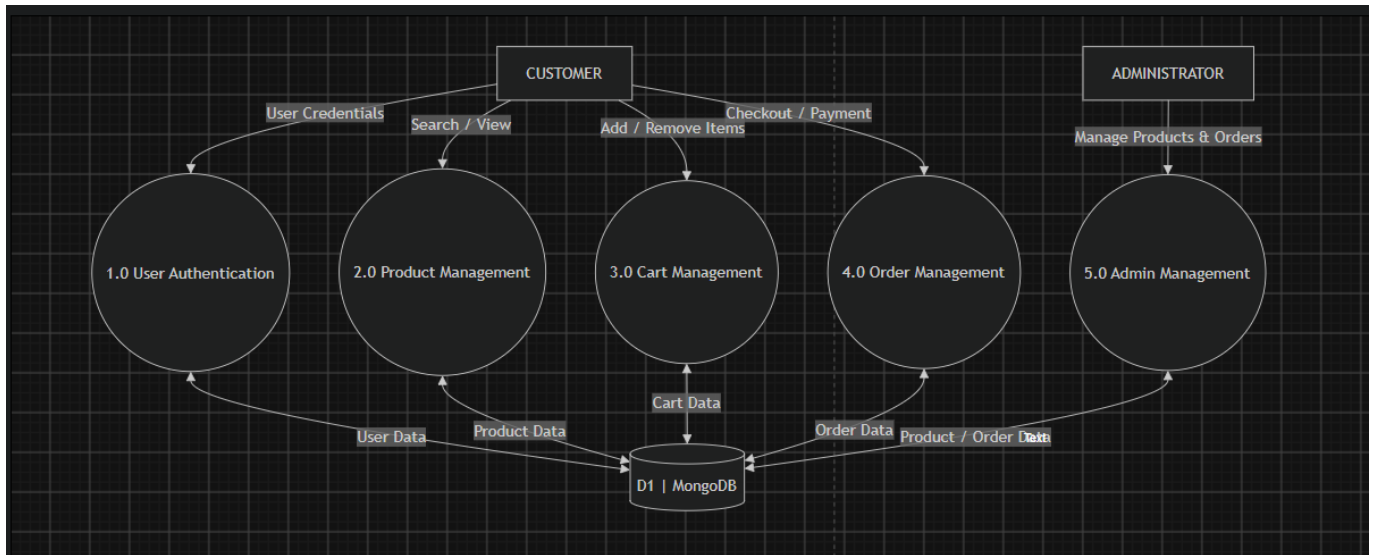
Data Flow Diagram — Level 0(Context Diagram):

Represents the entire ShopSphere system as a single core process (0.0). It maps top-level interactions between external entities (**Customer** and **Administrator**) and the platform.



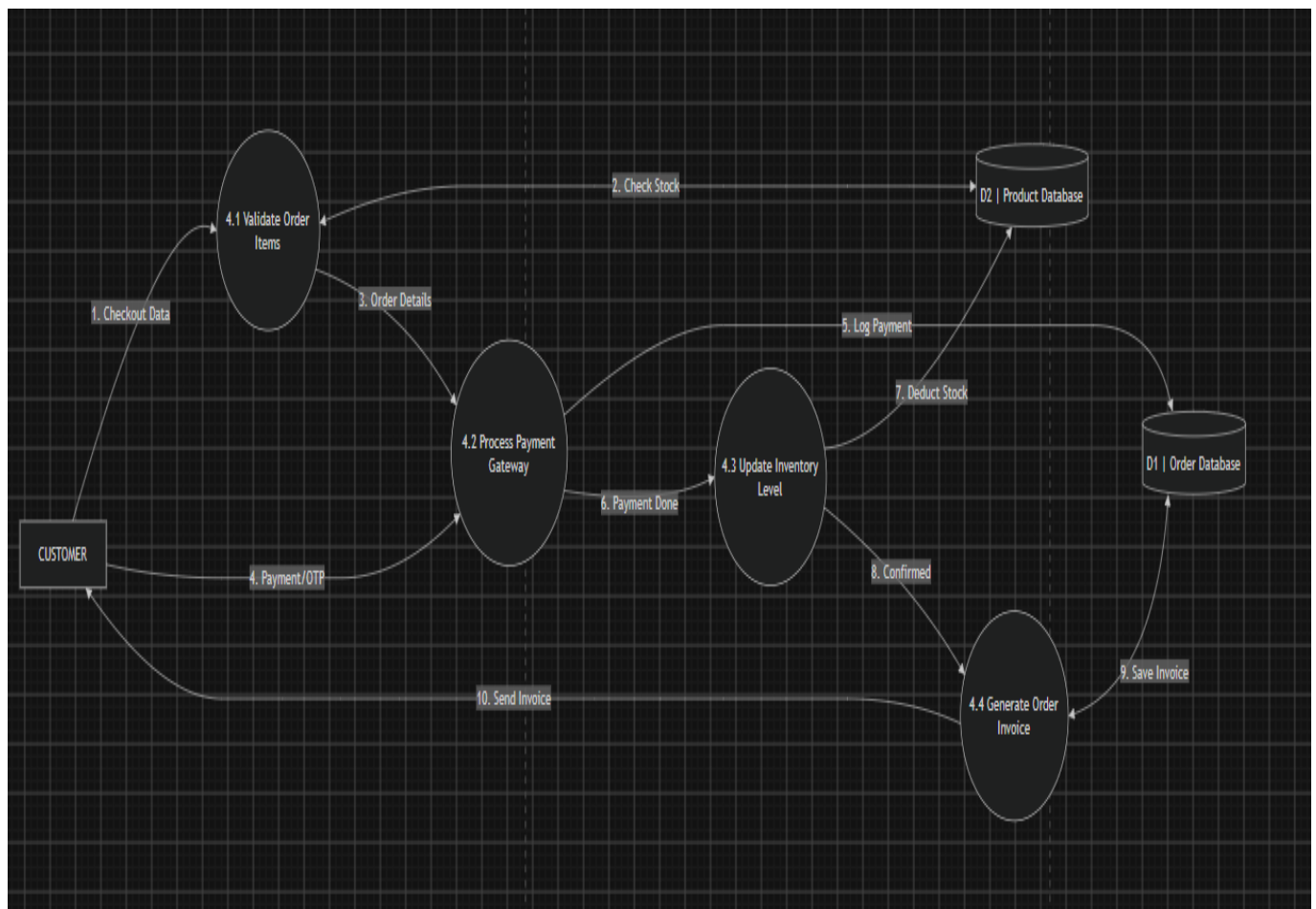
Data Flow Diagram — Level 1

- Decomposes the primary system process into five functional modules:
 - **1.0 User Authentication:** Manages registration, user login, and session validation.
 - **2.0 Product Browsing & Search:** Fetches real-time product catalogs and handles category filtering.
 - **3.0 Cart & Checkout Management:** Handles item additions, quantity modifications, and cart state persistence.
 - **4.0 Order & Payment Processing:** Processes user transactions, updates payment statuses, and generates order confirmations.
 - **5.0 Inventory & Admin Control:** Allows administrators to manage inventory, update product listings, and view sales reports.



Data Flow Diagram — Level 2

The Level 2 DFD expands the order workflow from product selection through order creation, storage and tracking.



SOURCES

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2. **Node.js Documentation** — Server-side JavaScript development.
3. **Express.js Documentation** — Backend services and REST APIs.
4. **MongoDB Documentation** — Database management and CRUD operations.
5. **Git Documentation** — Version control and project management.
6. **GitHub Documentation** — Repository hosting and collaboration.
7. **REST API Resources** — API design and HTTP communication.
8. **MDN Web Docs** — HTML, CSS, JavaScript, and web technologies.